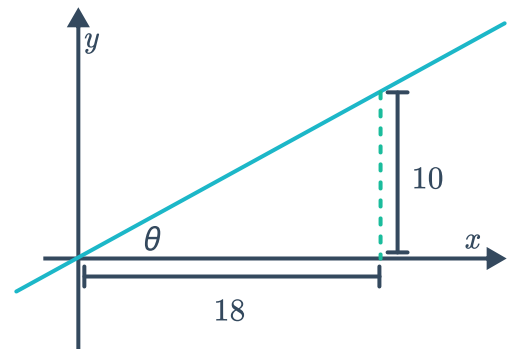


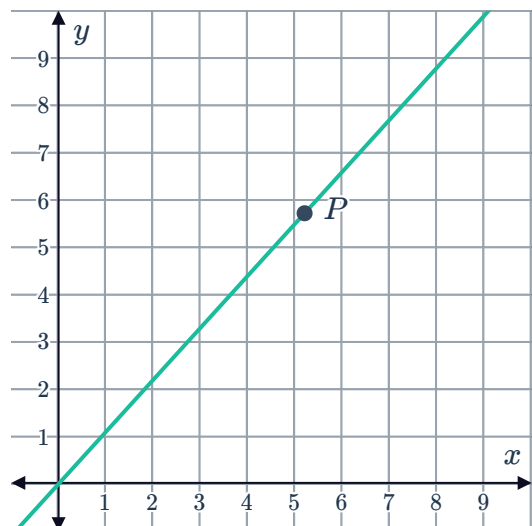
Gradient of a line



- 1 Write an expression for the gradient of the line that is inclined at an angle of 33° to the positive x -axis.
- 2 Find the gradient of a line that is inclined at an angle of 50° to the positive x -axis. Round your answer to one decimal place.
- 3 Find the angle of inclination of a line to the positive x -axis, whose gradient is 4. Round your answer to the nearest degree.
- 4 Consider the given line with an angle of inclination of θ with the positive x -axis:
 - a Find the gradient of the line.
 - b Hence, find θ . Round your answer to two decimal places.

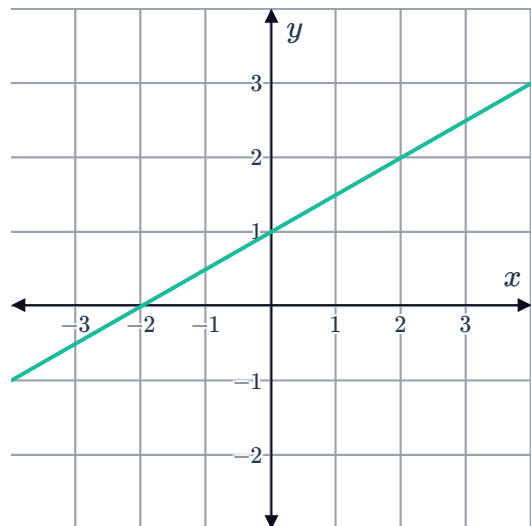


- 5 Point $P(5.2, 5.72)$ lies on the given line passing through the origin: Find θ , the angle of inclination of the line with the positive x -axis. Round your answer to two decimal places.



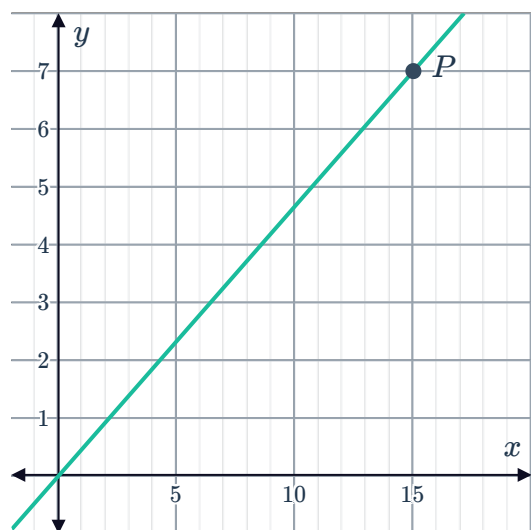
- 6 For each of the linear equations below, find θ the angle of inclination of the line with the positive x -axis. Round your answer to the nearest degree.
 - a $y = 6x + 2$
 - b $5x + 6y - 2 = 0$

- 7 Consider the graph of the line $y = \frac{1}{2}x + 1$ given and the triangle whose vertices are the origin, the x -intercept and the y -intercept. θ is the acute angle in the triangle that the line makes with the x -axis.



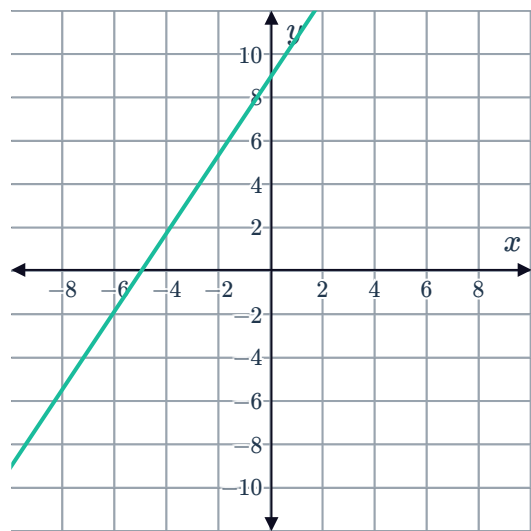
- a State the gradient of the line.
- b Find the value of $\tan \theta$.

- 8 Point $P(15, 7)$ lies on the given line passing through the origin: Find θ , the angle of inclination of the line with the positive x -axis. Round your answer to two decimal places.



- 9 A line makes an angle of θ with the positive x -axis. If $\theta < 90^\circ$. State whether the gradient of the line is positive or negative.
- 10 If a line has a gradient of -4 and an angle of inclination of θ with the positive x -axis, which of the following is true of the value of θ ?
- A** $-90^\circ < \theta < 0^\circ$ **B** $0^\circ < \theta < 90^\circ$ **C** $90^\circ < \theta < 180^\circ$ **D** $180^\circ < \theta < 270^\circ$
- 11 Two points $A(1, -2)$ and $B(9, 30)$ lie on a line that makes an angle of θ degrees with the positive x -axis.
- a Find the gradient of the line.
 - b Find θ to two decimal places.

- 12 Find θ , the angle of inclination of the line with the positive x -axis, if the x -intercept is -5 and the y -intercept is 9 . Round your answer to two decimal places.



- 13 A line with gradient m has an angle of inclination of 65° with the positive x -axis, such that $\tan 65^\circ = m$. Without solving for m , find the value of θ such that $\tan \theta = -\frac{1}{m}$.
- 14 Find θ , the angle of inclination of the line with the positive x -axis for a line with a gradient of $-\frac{1}{\sqrt{3}}$.
- 15 A horizontal line makes an angle of 0° with the x -axis.
- Evaluate $\tan 0^\circ$ by considering its relationship to the gradient of the line.
 - A vertical line is perpendicular to a horizontal line. Using the fact that the gradients of perpendicular lines have the relationship $m_1 \times m_2 = -1$, find the value of $\tan 90^\circ$.
- 16 A line is drawn on the cartesian plane at an angle of inclination θ with the positive x -axis. Given that $\sin \theta = \frac{12}{13}$, find the gradient m of the line.

Applications

- 17 Line L_1 makes an angle of θ with the positive x -axis. Line L_2 is such that it is perpendicular to line L_1 . Form an expression for the gradient of line L_2 in terms of θ .
- 18 A line passing through the points $(2, -3)$ and $(x, 9)$ makes an angle of 120° with the positive x -axis. Find the exact value of x .

19 The steepness of a ski run is defined by the angle of inclination of the run with the horizontal ground.



By how many degrees is Run A steeper than Run B? Round your answer to one decimal place.

